Adrenaline

Mechanism of action

- Adrenaline stimulates alpha and beta receptors, with the predominant effects occurring at alpha 1, beta 1 and beta 2 receptors.
- Alpha 1 stimulation causes smooth muscle contraction, vasoconstriction of blood vessels and stimulation of glycogenolysis and gluconeogenesis.
- Beta 1 stimulation causes an increase in inotropy (cardiac contractility), an increase in chronotropy (heart rate) and an increase in dromotropy (speed of electrical conduction within the heart).
- Beta 2 stimulation causes smooth muscle relaxation, skeletal muscle vasodilation, bronchodilation, and stabilisation of mast cell membranes, reducing histamine release.

Authorised practice levels

- EMTs: nebulised, IM, IN and topical adrenaline.
- Paramedics: all of the above and adrenaline IV for cardiac arrest.
- ICPs and CCPs: all indications and all routes.

Indications

- Cardiac arrest.
- Anaphylaxis.
- Severe asthma.
- Imminent respiratory arrest from COPD.
- Bradycardia with moderate or severe cardiovascular compromise.
- Blood pressure support if unresponsive to metaraminol.
- Septic shock, cardiogenic shock, and neurogenic shock, unresponsive to 0.9% sodium chloride IV and metaraminol IV.
- Stridor causing moderate or severe respiratory distress.
- IN for clinically significant epistaxis.
- Topical for clinically significant bleeding from a wound.

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Contraindications

- ✕ None.

Cautions

- ⚠ Myocardial ischaemia. Adrenaline will increase myocardial oxygen consumption.
- ⚠ Tachydysrhythmias. Adrenaline will usually make tachydysrhythmias worse.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

Dosage

- The dose of adrenaline is dependent on the indication and the route. See the individual guideline.

Administration

- Topical: dilute each mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Apply topically in addition to direct pressure.
- IN: dilute each mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Administer the appropriate dose into each bleeding nostril using a mucosal atomising device, in addition to direct pressure.
- Nebulised: administer undiluted.
- IM: administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.
- Cardiac arrest:
  - » Adults and children whose weight has been rounded to 50 kg or more: administer undiluted as an IV bolus.
  - » Children whose weight has been rounded to 40 kg or less: dilute 1 mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Draw up the dose from this solution and administer as an IV bolus.

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- IV infusion: place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride. Shake well and label. This solution is 1:1,000,000 and contains 0.001 mg/ml.
  - For an adult: administer as an IV infusion starting at 2 drops/second. Adjust the rate to the patient's condition.
  - For a child aged 5-14 years: administer as an IV infusion starting at 1 drop/second. Adjust the rate to the patient's condition.
- When administering an IV infusion of adrenaline using 1 mg of adrenaline in a 1 litre bag of 0.9% sodium chloride:
  - 2 drops/second via a standard IV administration set will administer approximately 0.4 mg/hour of adrenaline.
  - Record an estimate of the total dose of adrenaline administered.
- For all other IV administration: place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride, Shake well and label. This solution is 1:1,000,000 and contains 0.001 mg/ml. Draw up the dose from this solution and administer as an IV bolus.

Common adverse effects

- Tachycardia.
- Tachydysrhythmia.
- Myocardial ischaemia.
- Ventricular ectopy.
- Hypertension.
- Nausea and vomiting.
- Tremor, anxiety and sweating.
- Hyperglycaemia.

Usual onset of effect

- IV: 5-10 seconds.
- IM: 2-5 minutes.
- Nebulised, IN and topical: on contact with the target site.

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Usual duration of effect

- The cardiovascular effects last 5-15 minutes.
- The mast cell membrane effects may last for several hours.

Usual preparation

- Ampoule containing 1 mg in 1 ml.

Pharmacokinetics

- Adrenaline is metabolised by the liver and taken up by sympathetic nerve endings.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Increased doses may be required if the patient is taking a beta-blocker or a calcium channel blocker. This effect is particularly prominent in the setting of poisoning if a large dose of a beta-blocker and/or calcium channel blocker has been taken.

More Information

Version history

22 November 2022 1.0.1
Update: Indication to align with stridor and croup CPGs.